SDITO LAB

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Assignment 07

Problem Statement : Hosting a website on EC2

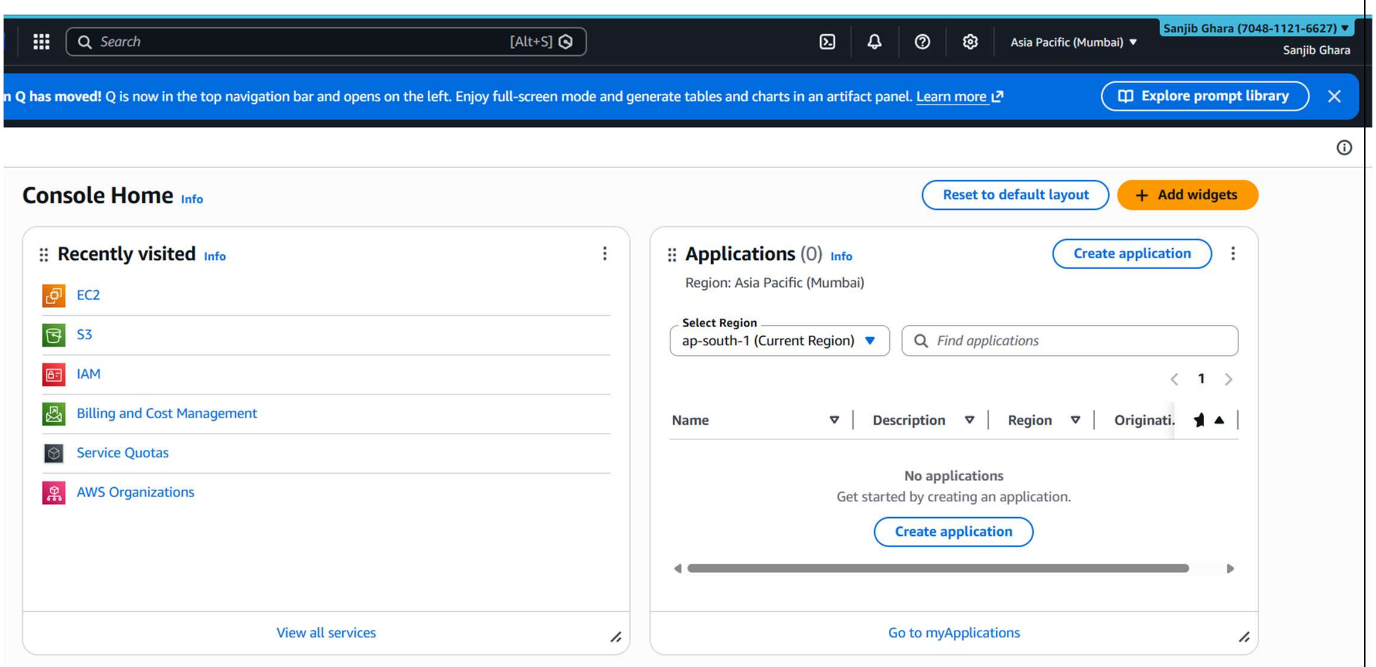
Aim
To launch an Amazon EC2 instance, configure a web server, and host a static website using terminal-based configuration.

Requirements
- AWS Account
- Key pair (.pem file)
- HTML file (index.html)
- Internet connection
- Bitvise SSH Client

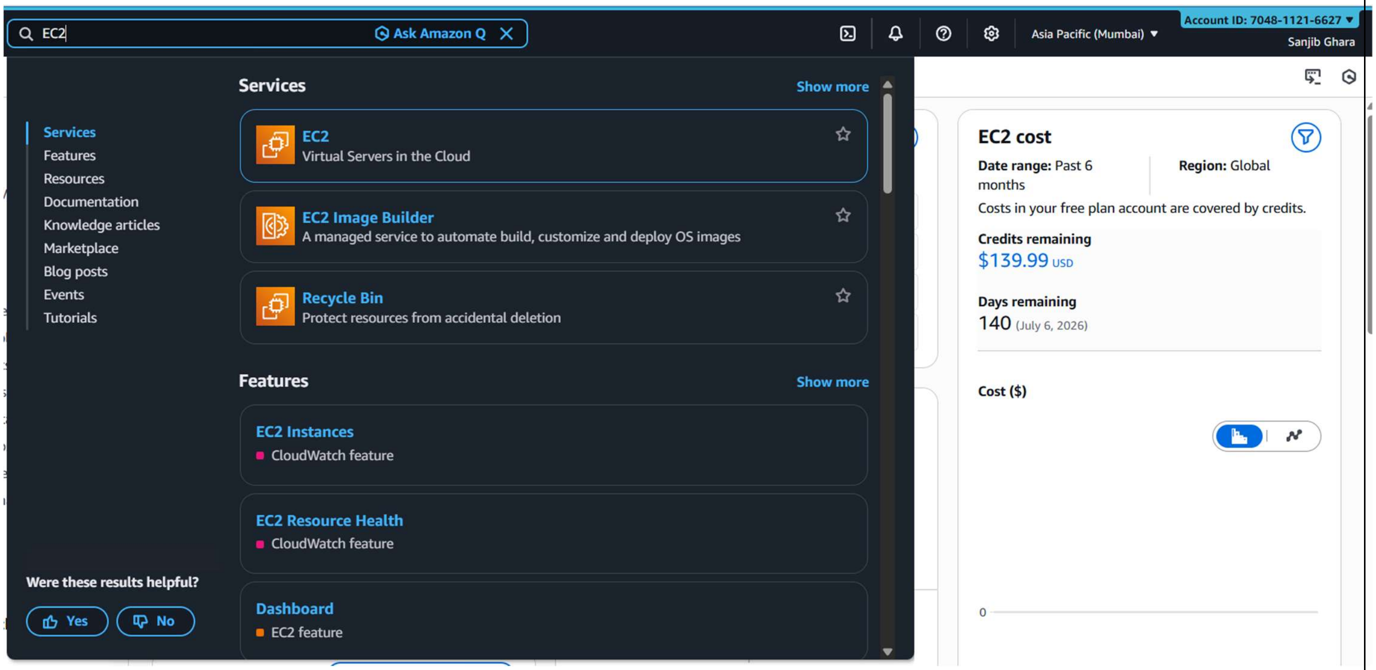
Solution:

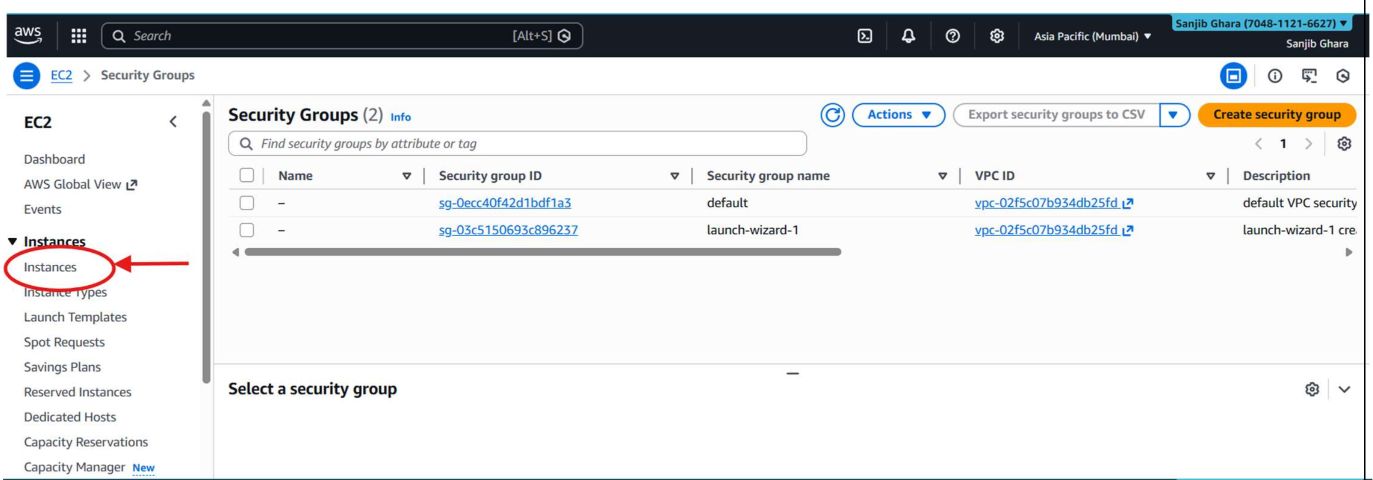
Step 1: Launch an EC2 Instance

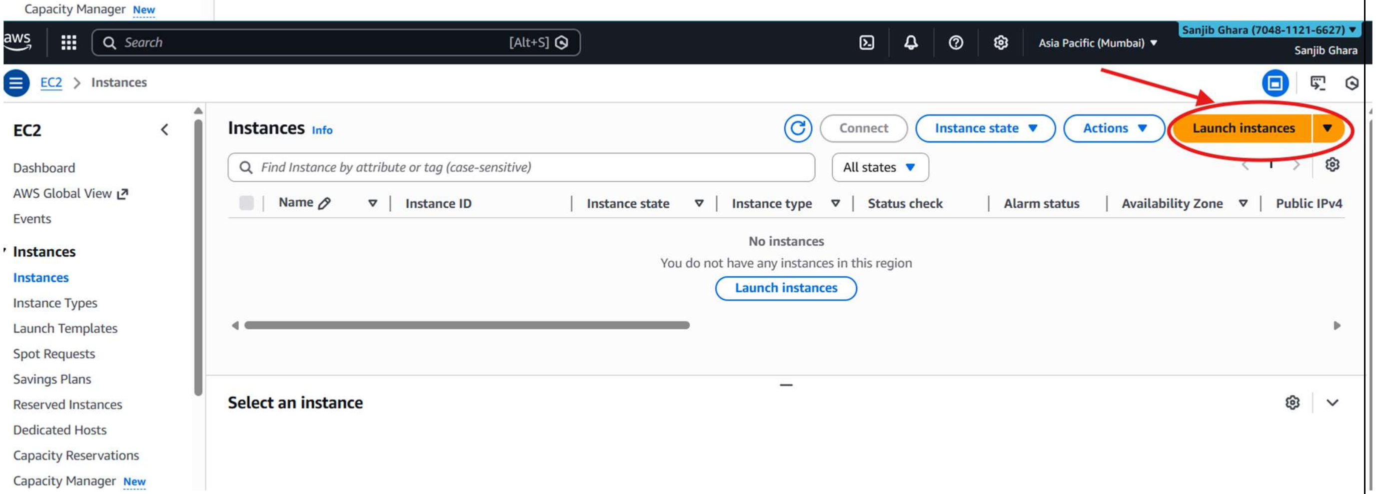
1. Login to AWS Management Console.



2. Search for EC2 and open the service.



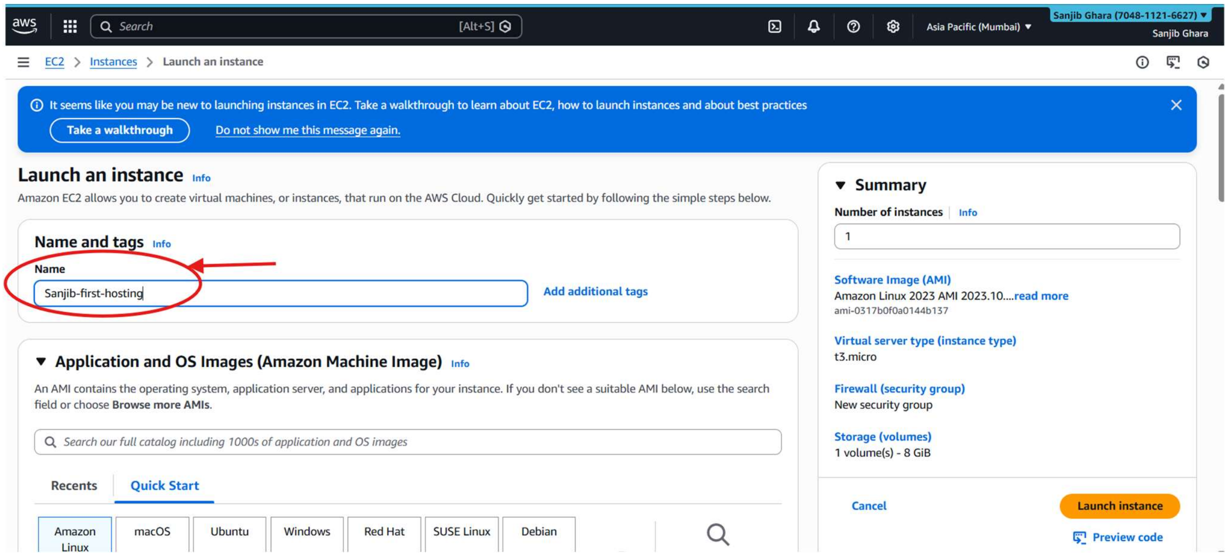


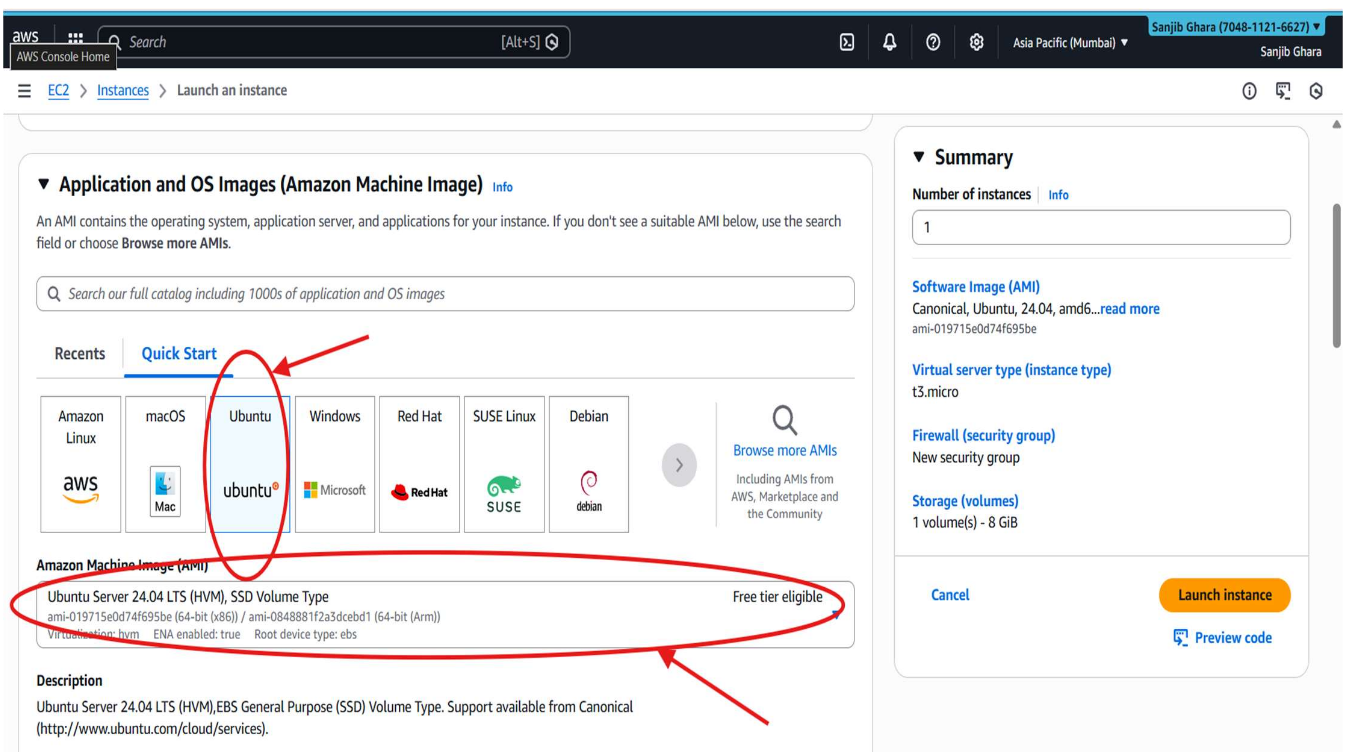


3. Click Launch instance.

<u>Instance Configuration</u>

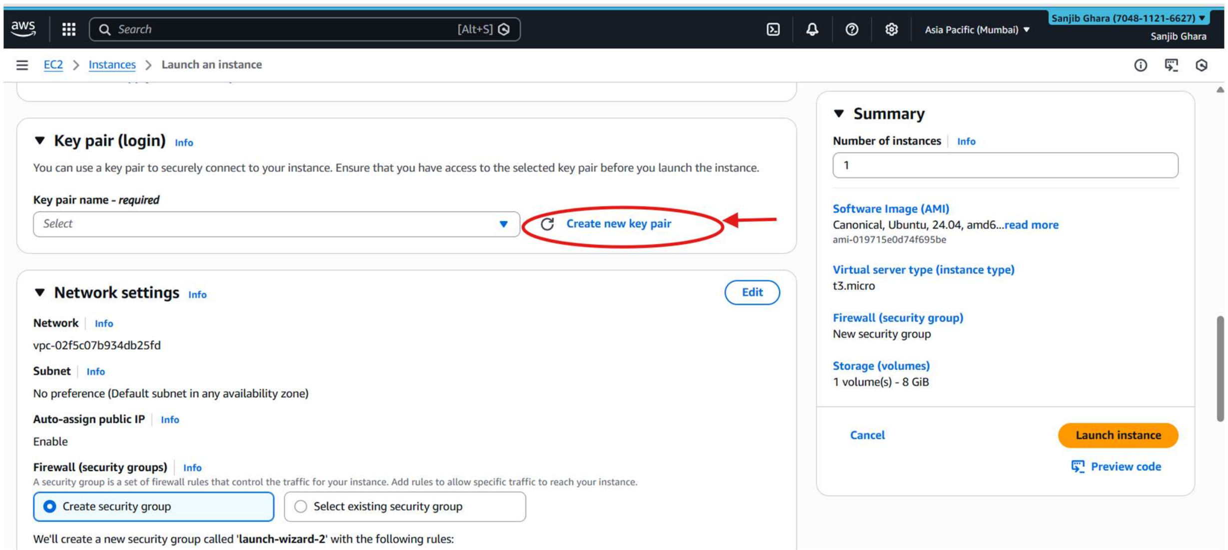
4. **Enter:**
    - Name: Sanjib-first-hosting
5. **Select:**
    - AMI: Ubuntu (Free Tier Eligible)

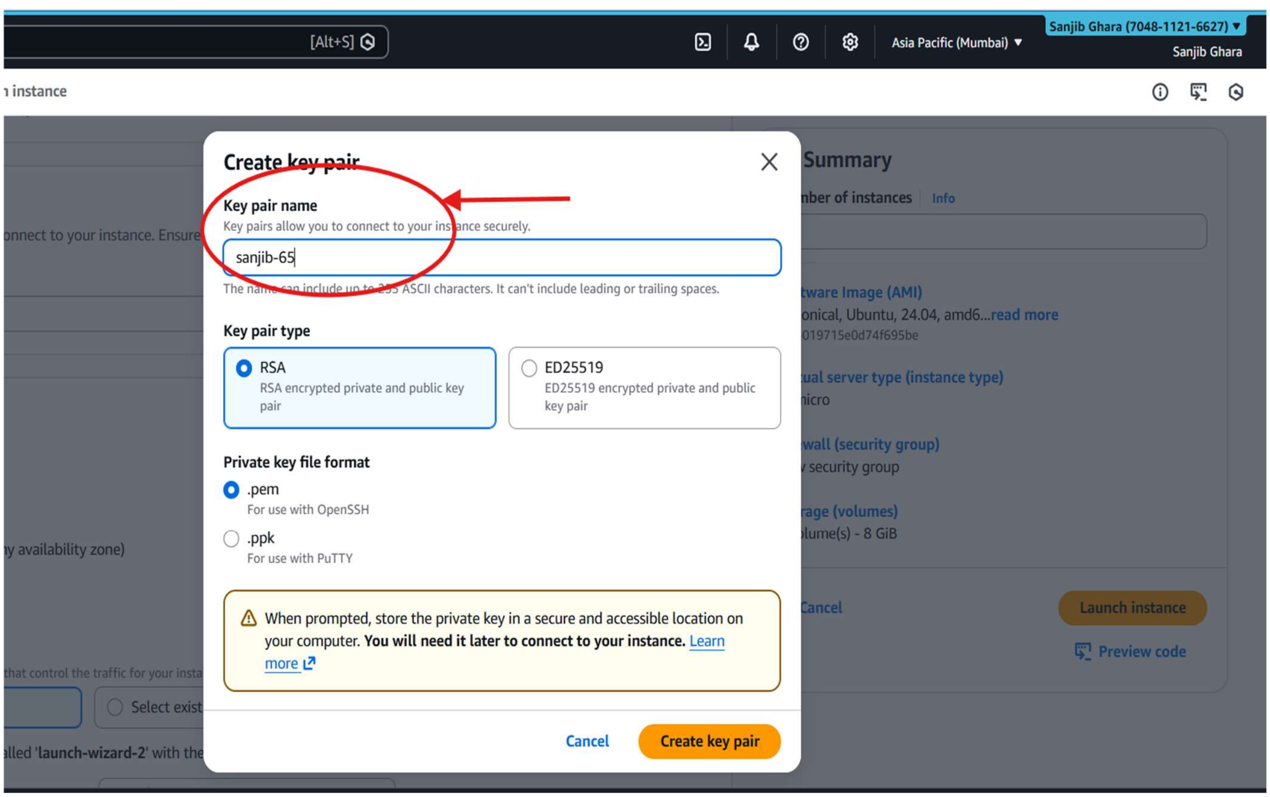




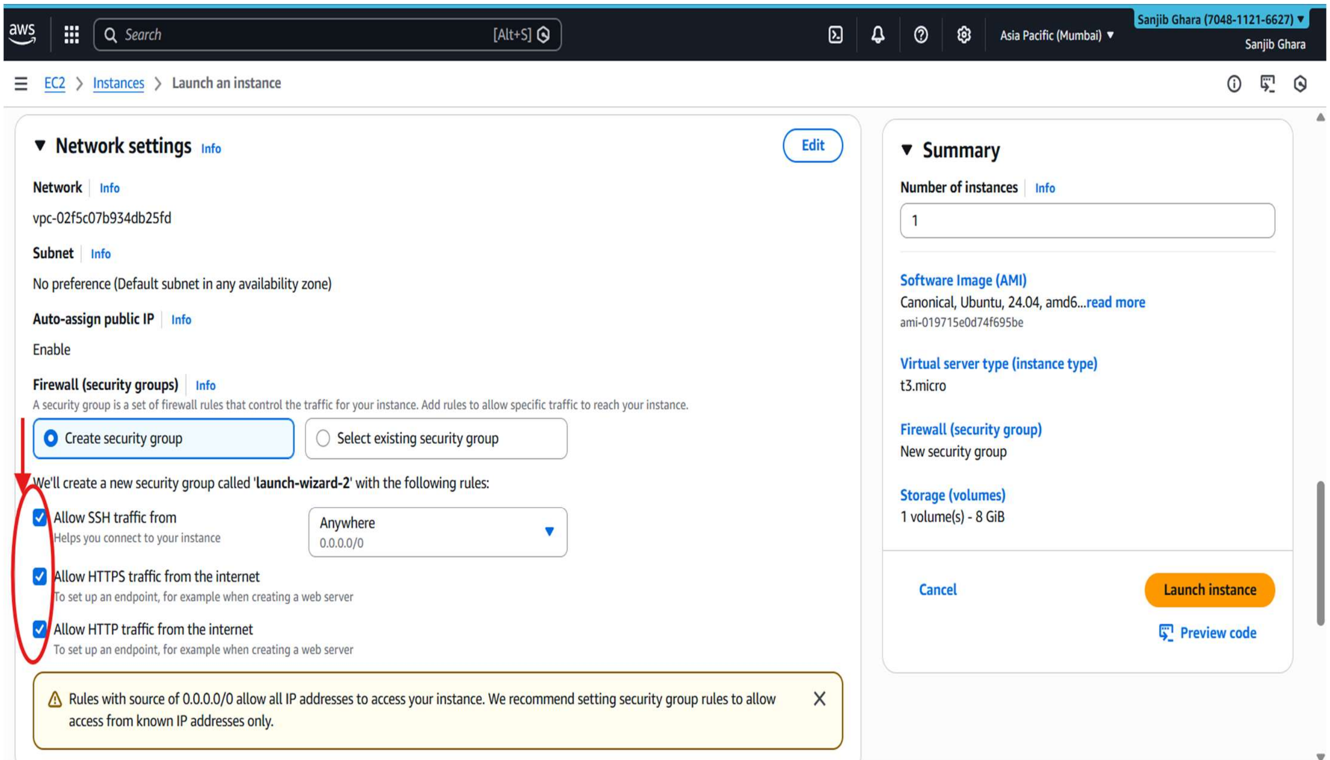
6. Click Create new key pair:

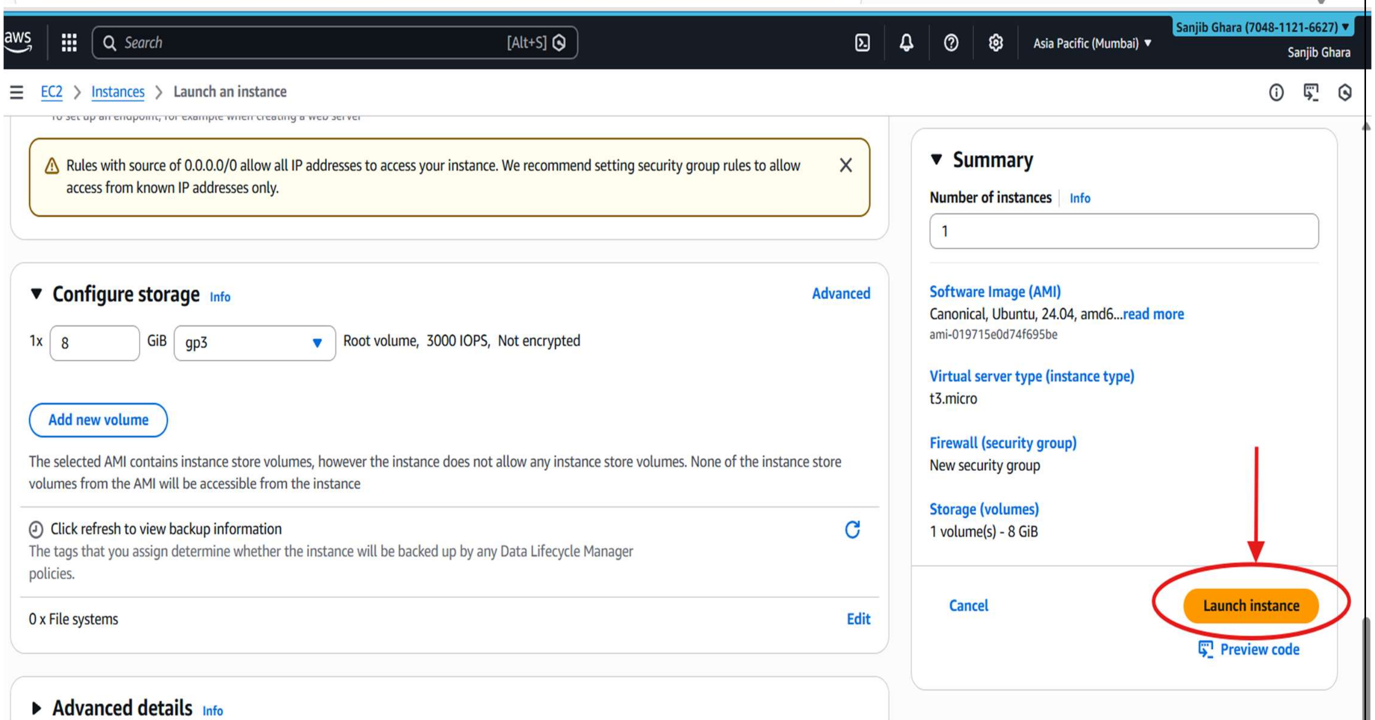
- Name: Sanjib-65
- Type: RSA
- Format: .pem



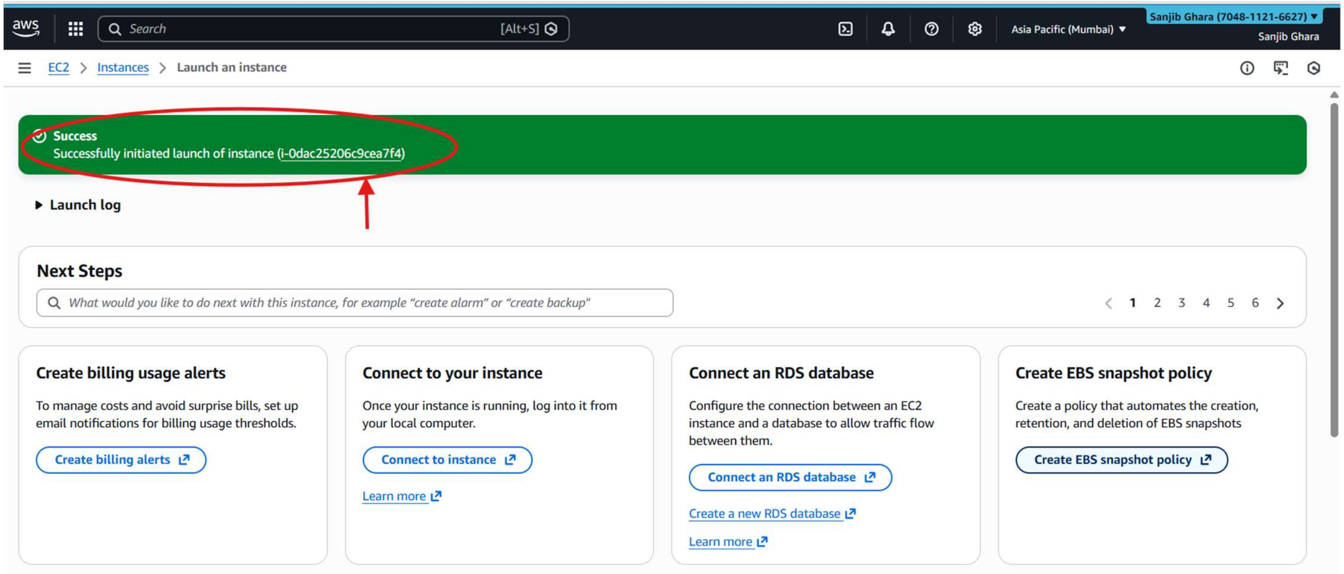


7. Download and save the key file securely.
8. Click on **Create security group** and check the boxes of **Allow HTTPS traffic from the internet and Allow HTTP traffic from the internet** and click on **Launch instance** button.

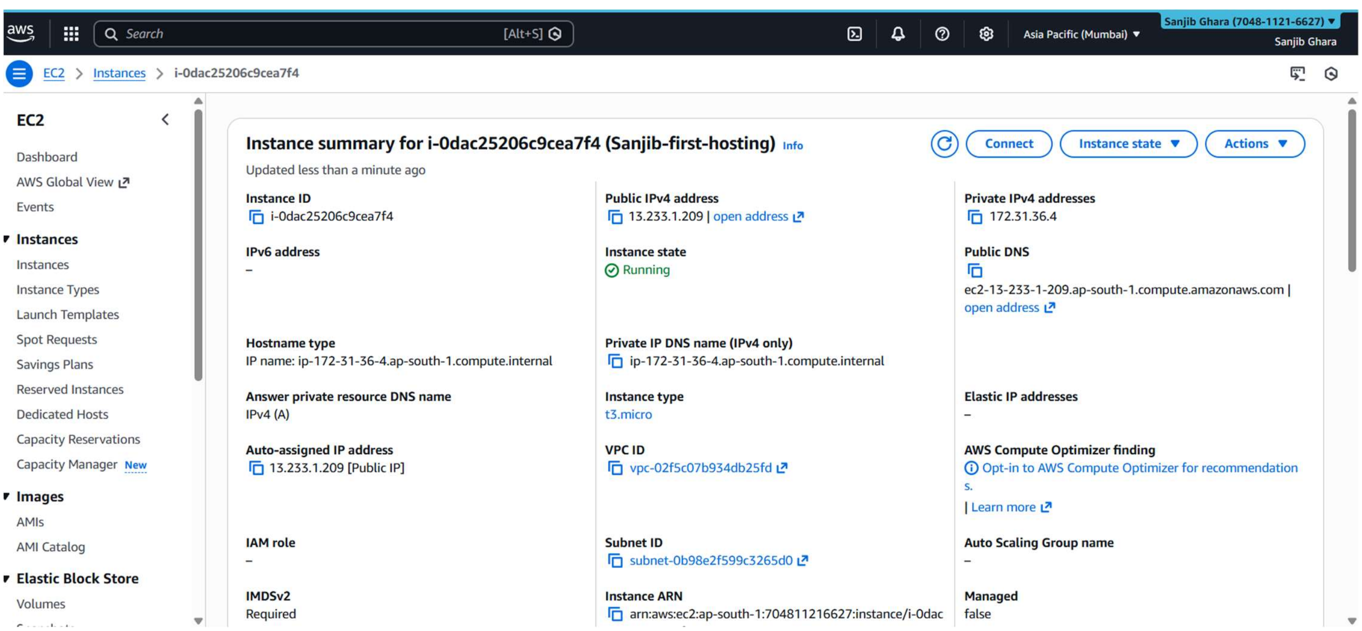




9. The Instance will be successfully created , return to the **Instance** page.



10: The Instance will be seen running, click on the newly created instance



Step 2: Copy Public IPv4 Address

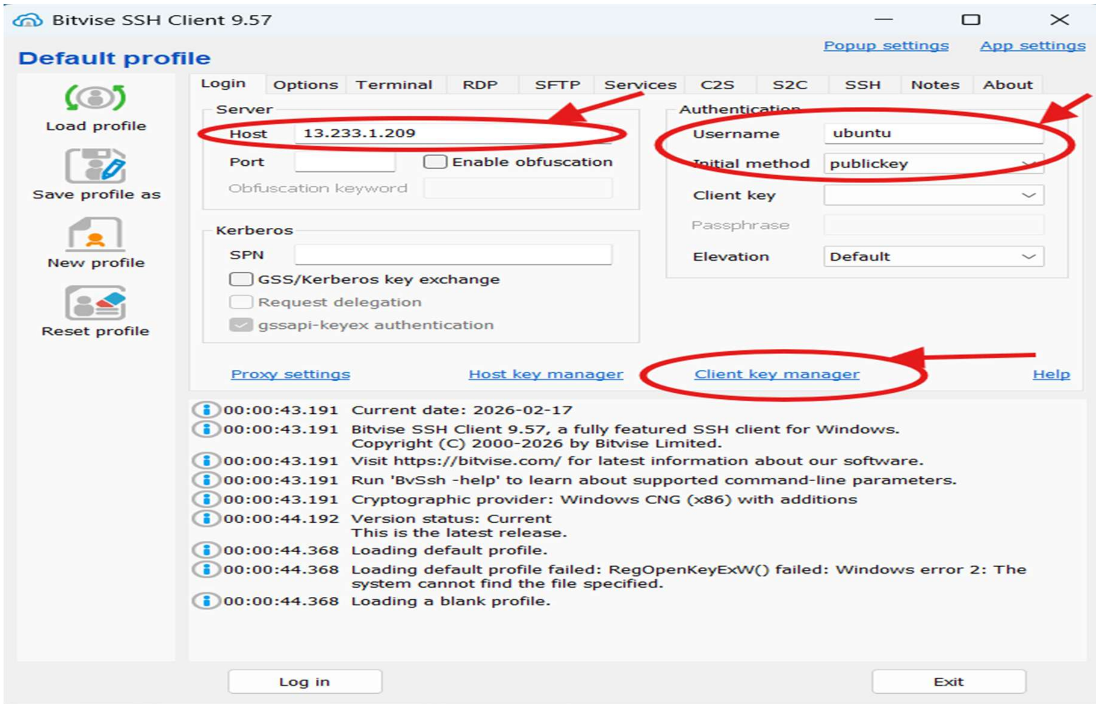
After the EC2 instance is launched and running, copy the Public IPv4 address from the Instance details page.

Step 3: Open Bitvise SSH Client and Configure Connection

1. Open Bitvise SSH Client on the computer.
2. Paste the copied Public IPv4 address into the Host field.
3. Enter ubuntu in the Username field.

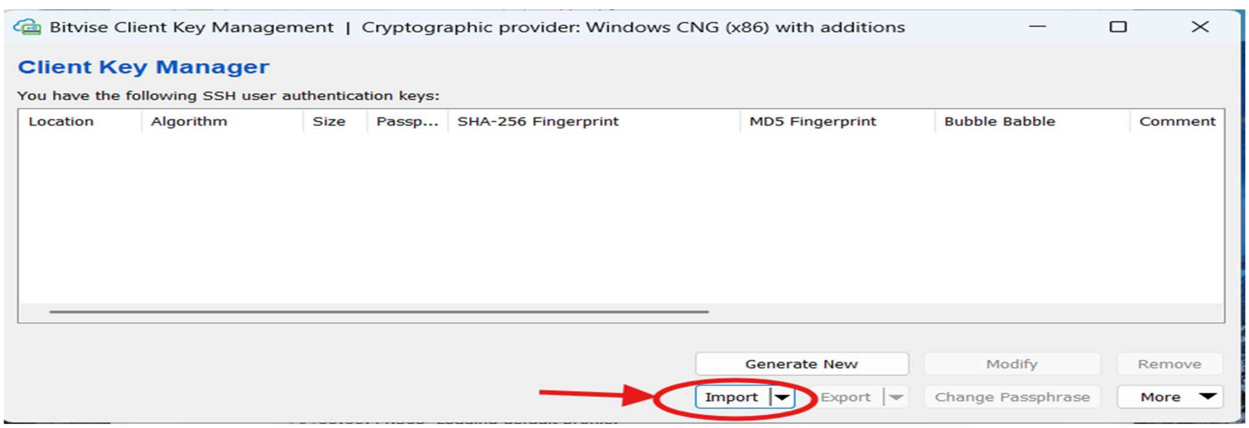
4.Set Initial method to publickey.

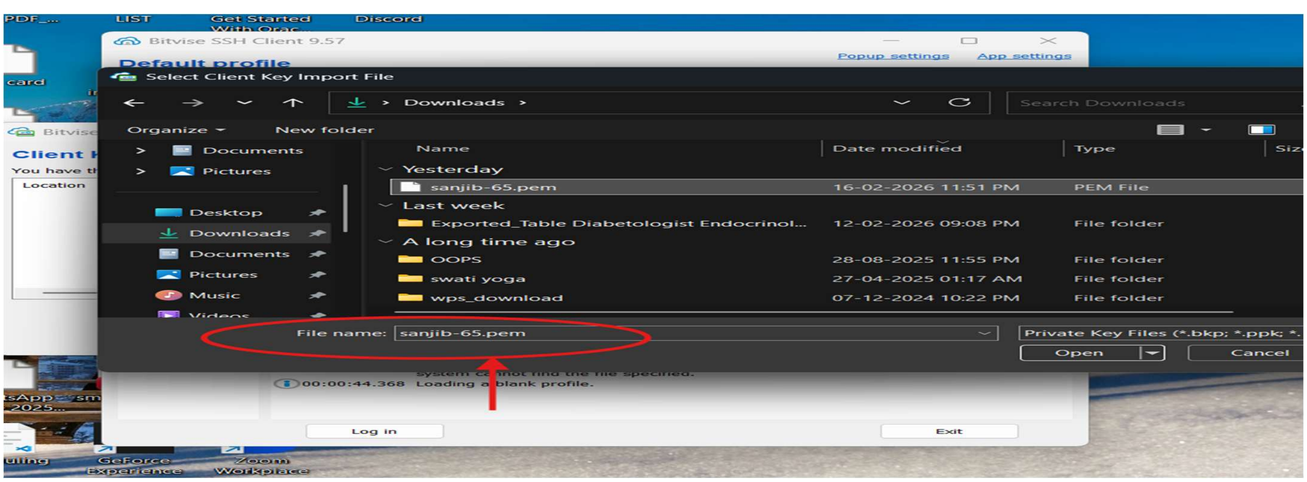
5. Click on Client key manager

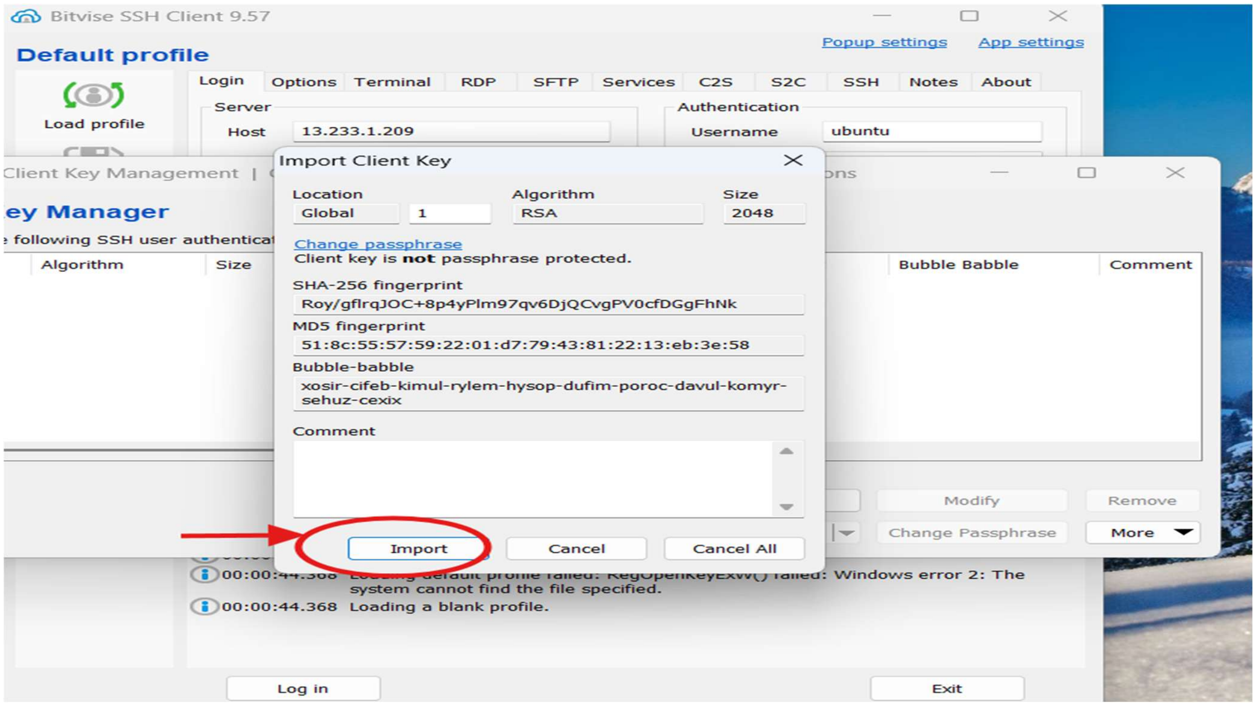


Step 4: Import Private Key

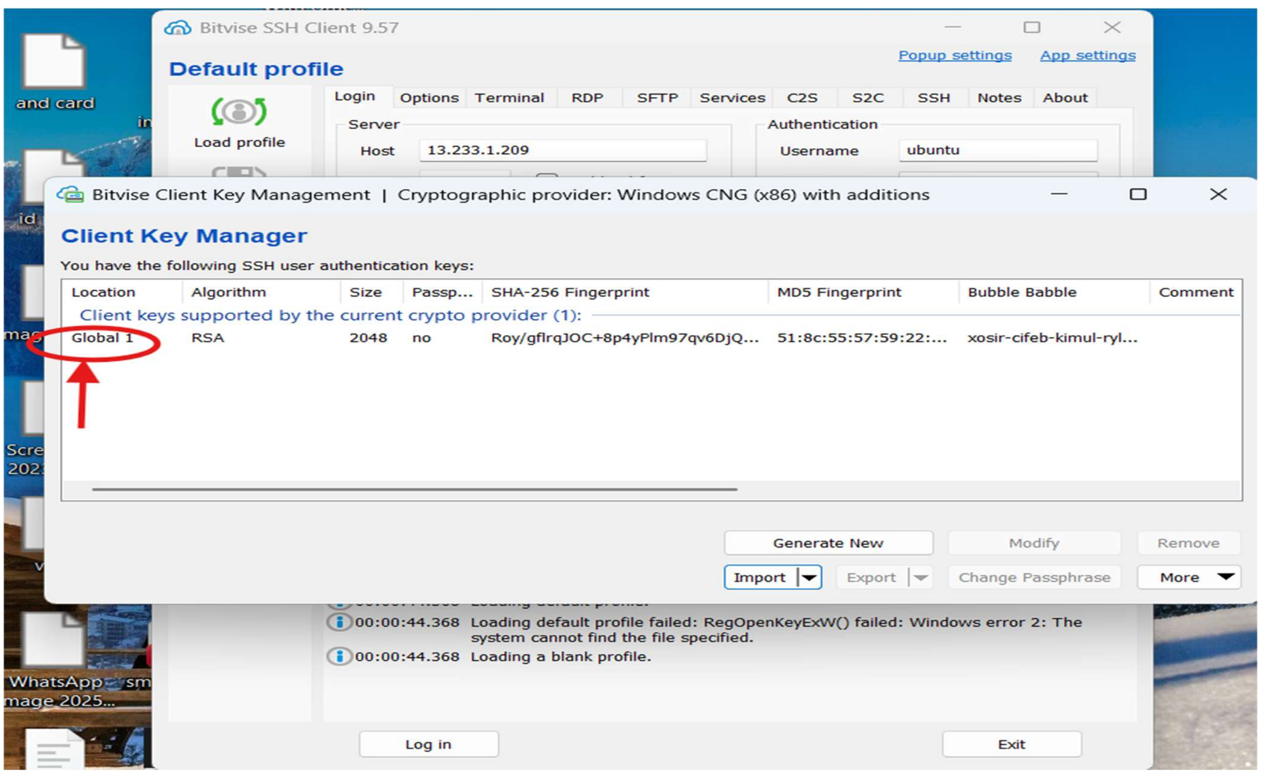
1. In Client Key Manager, click Import.
2. Select the downloaded .pem key file from the Desktop.





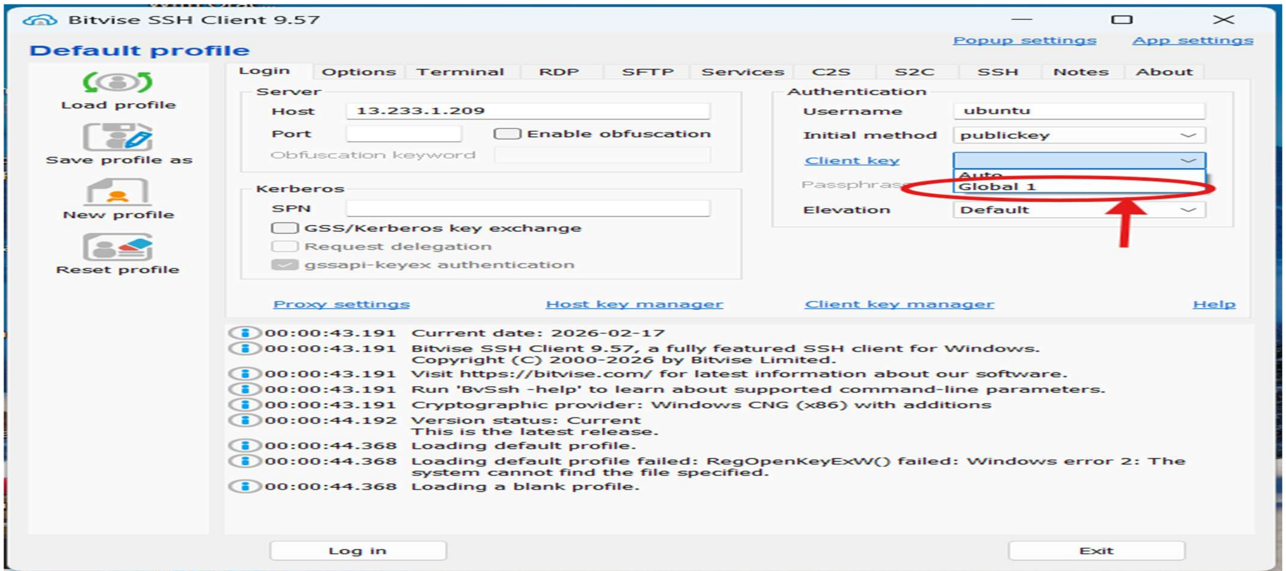


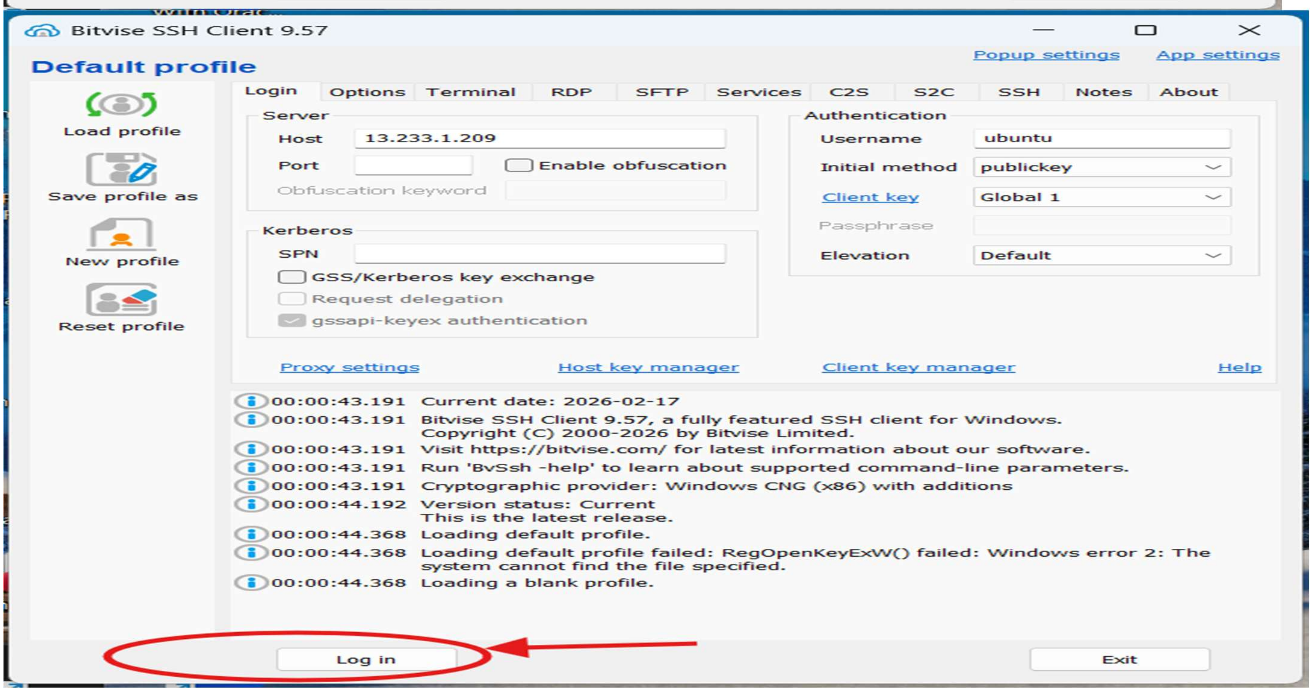
3.Save the key.



Step 5: Login to the Instance

1. Ensure Client key is set to Global 1.
2. Click on the Log in button.





Step 6: Open Terminal Console

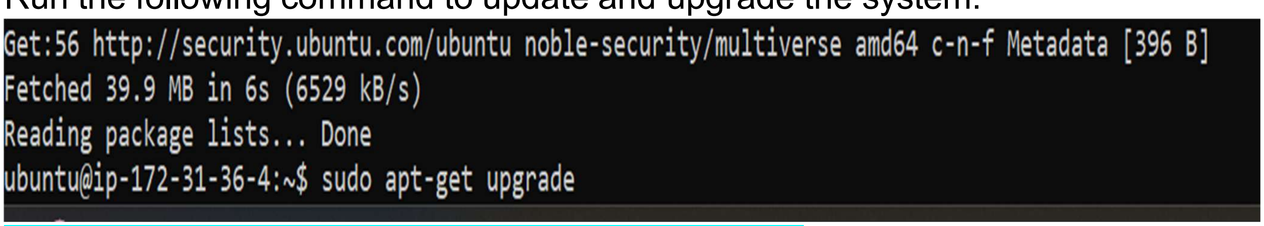
After successful login:

- Click on **New Terminal Console** from the **Bitvise** side panel.

-

Step 7: Update Ubuntu System

Run the following command to update and upgrade the system:



sudo apt-get update && sudo apt-get upgrade -y

Step 8: Install Nginx Web Server

Install Nginx using the command:

```
User sessions running outdated binaries:
 ubuntu @ session #2: sshd[1061,1172]
 ubuntu @ user manager service: systemd[1066]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-36-4:~$ sudo apt-get install nginx
```

sudo apt-get install nginx -y

```
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Step 9: Set Permissions for Web Directory

Navigate to the web directory:

cd /var/www

```
ubuntu@ip-172-31-36-4:/home$ cd..
cd..: command not found
ubuntu@ip-172-31-36-4:/home$ cd ..
ubuntu@ip-172-31-36-4:/$ ls
bin                boot  etc   lib                 lib64       media  opt   root  sbin                snap  sys  usr
bin.usr-is-merged  dev   home  lib.usr-is-merged   lost+found  mnt    proc  run   sbin.usr-is-merged  srv   tmp  var
ubuntu@ip-172-31-36-4:/$
```

```
ubuntu@ip-172-31-36-4:/$ cd var
ubuntu@ip-172-31-36-4:/var$
```

```
ubuntu@ip-172-31-36-4:/var$ ls
backups  cache  crash  lib  local  lock  log  mail  opt  run  snap  spool  tmp  www
ubuntu@ip-172-31-36-4:/var$
```

```
ubuntu@ip-172-31-36-4:/var$ cd www/
ubuntu@ip-172-31-36-4:/var/www$ ls
html
ubuntu@ip-172-31-36-4:/var/www$
```

Give proper permissions to the HTML directory:

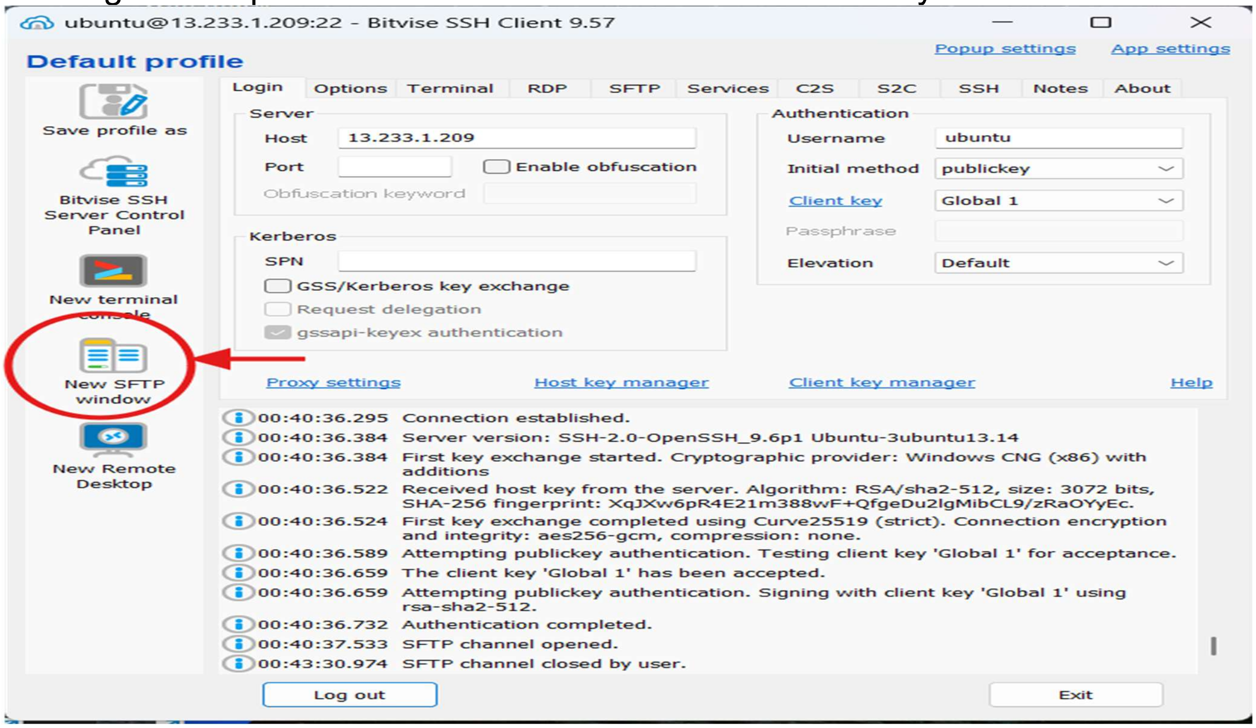
sudo chmod 777 html/

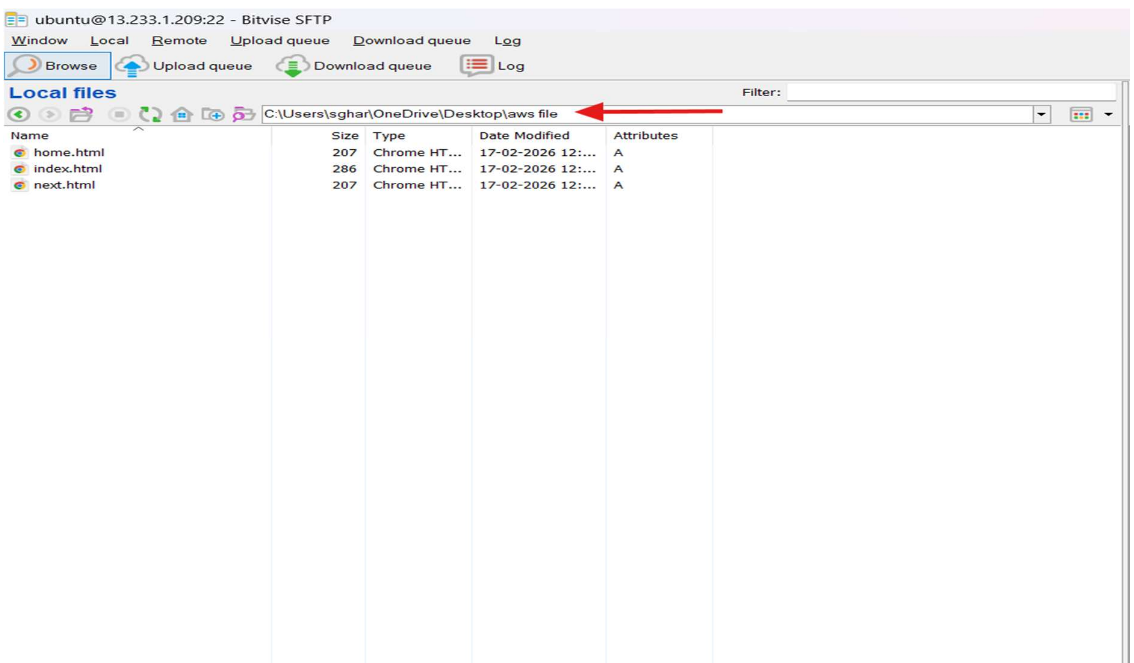
```
ubuntu@ip-172-31-36-4:/var/www$ ls
html
ubuntu@ip-172-31-36-4:/var/www$ sudo chmod 777 html
ubuntu@ip-172-31-36-4:/var/www$
```

Step 10: Upload Website Files Using SFTP

1. Click New SFTP window in Bitvise.
2. In the left pane (Local Files), navigate to the folder containing:
    - index.html
    - prev.html
    - next.html

3. In the right pane (Remote Files), navigate to: **/var/www/html**
4. Drag and drop all three files from local to remote directory.





Remote files

Filter:

/home/ubuntu

Name	Size	Type	Date Modified	Permissions	Owner	Group
.cache	4,096	File folder	17-02-2026 12:...	drwx------	ubuntu	ubuntu
.ssh	4,096	File folder	16-02-2026 11:...	drwx------	ubuntu	ubuntu
.bash_history	139	BASH_HIST...	17-02-2026 12:...	-rw-------	ubuntu	ubuntu
.bash_logout	220	Bash Logou...	31-03-2024 02:...	-rw-r--r--	ubuntu	ubuntu
.bashrc	3,771	Bash RC So...	31-03-2024 02:...	-rw-r--r--	ubuntu	ubuntu
.profile	807	Profile Sou...	31-03-2024 02:...	-rw-r--r--	ubuntu	ubuntu
.sudo_as_admin_successful	0	SUDO_AS_...	17-02-2026 12:...	-rw-r--r--	ubuntu	ubuntu

Remote files

Filter:

/home

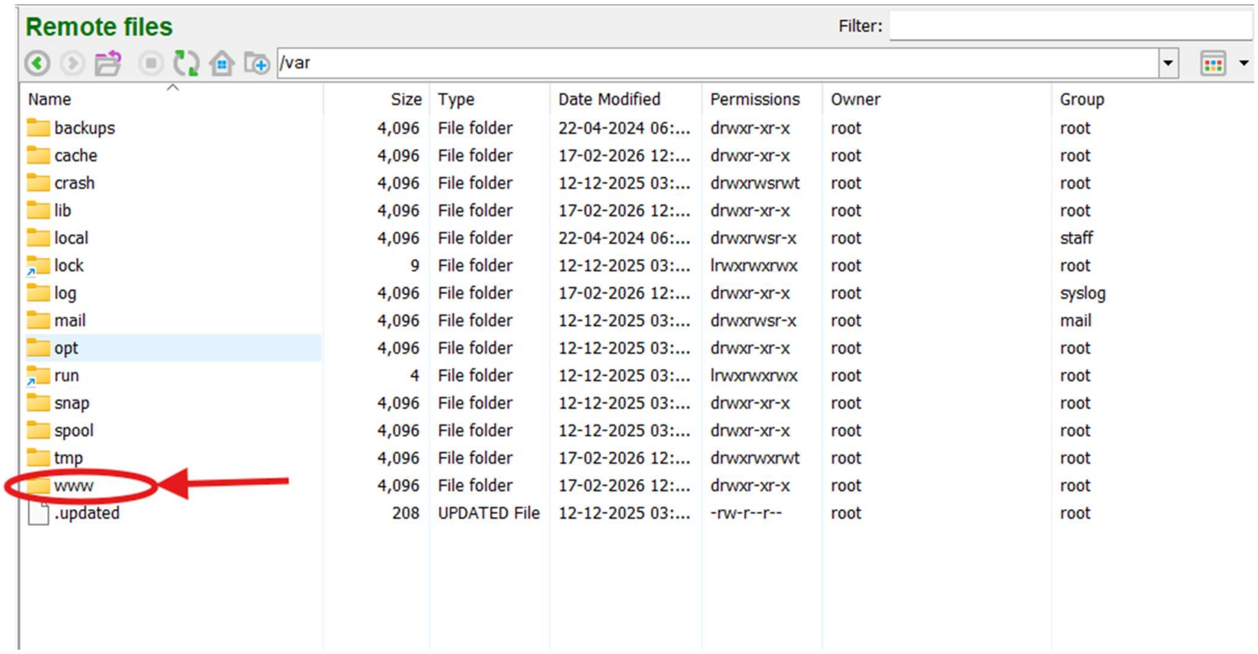
Name	Size	Type	Date Modified	Permissions	Owner	Group
ubuntu	4,096	File folder	17-02-2026 12:...	drwxr-x---	ubuntu	ubuntu

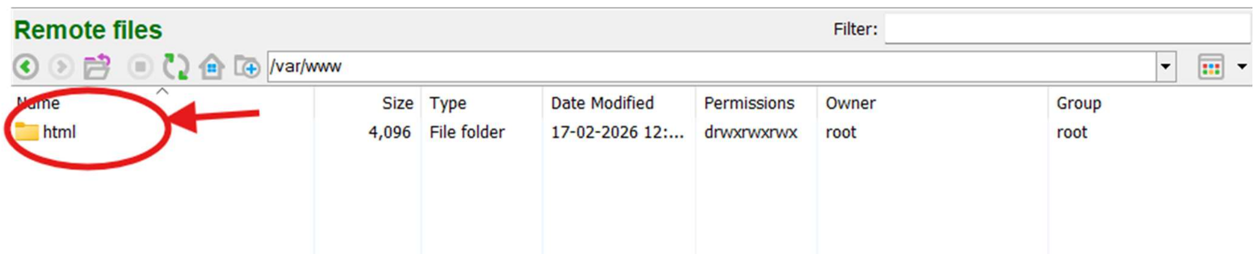
Remote files

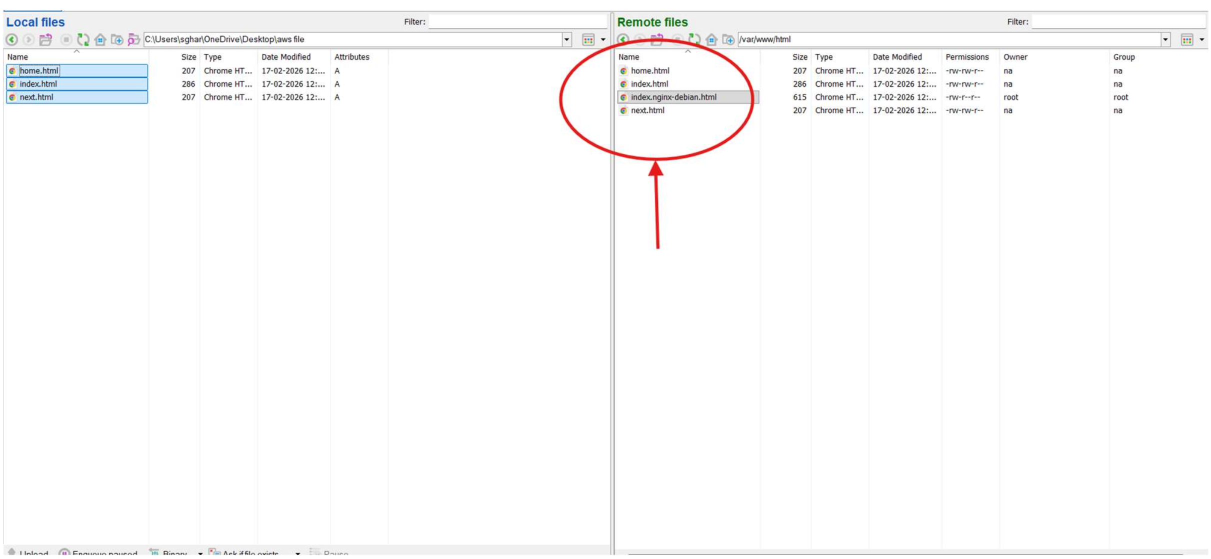
Filter:

/

Name	Size	Type	Date Modified	Permissions	Owner	Group
bin	7	File folder	22-04-2024 06:...	lrwxrwxrwx	root	root
bin.usr-is-merged	4,096	File folder	26-02-2024 06:...	drwxr-xr-x	root	root
boot	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	root
dev	3,300	File folder	16-02-2026 11:...	drwxr-xr-x	root	root
etc	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	root
home	4,096	File folder	16-02-2026 11:...	drwxr-xr-x	root	root
lib	7	File folder	22-04-2024 06:...	lrwxrwxrwx	root	root
lib.usr-is-merged	4,096	File folder	08-04-2024 08:...	drwxr-xr-x	root	root
lib64	9	File folder	22-04-2024 06:...	lrwxrwxrwx	root	root
lost+found	16,384	File folder	12-12-2025 03:...	drwx------	root	root
media	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
mnt	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
opt	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
proc	0	File folder	16-02-2026 11:...	dr-xr-xr-x	root	root
root	4,096	File folder	16-02-2026 11:...	drwx------	root	root
run	1,020	File folder	17-02-2026 12:...	drwxr-xr-x	root	root
sbin	8	File folder	22-04-2024 06:...	lrwxrwxrwx	root	root
sbin.usr-is-merged	4,096	File folder	31-03-2024 02:...	drwxr-xr-x	root	root
snap	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
srv	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
sys	0	File folder	16-02-2026 11:...	dr-xr-xr-x	root	root
tmp	4,096	File folder	17-02-2026 12:...	drwxrwxrwt	root	root
usr	4,096	File folder	12-12-2025 03:...	drwxr-xr-x	root	root
var	4,096	File folder	17-02-2026 12:...	drwxr-xr-x	root	root


Remote files
Filter:
/var
Name | Size | Type | Date Modified | Permissions | Owner | Group
backups 4,096 File folder 22-04-2024 06:... drwxr-xr-x root root
cache 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
crash 4,096 File folder 12-12-2025 03:... drwxrwsrwt root root
lib 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
local 4,096 File folder 22-04-2024 06:... drwxrwsr-x root staff
lock 9 File folder 12-12-2025 03:... lrwxrwxrwx root root
log 4,096 File folder 17-02-2026 12:... drwxr-xr-x root syslog
mail 4,096 File folder 12-12-2025 03:... drwxrwsr-x root mail
opt 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
run 4 File folder 12-12-2025 03:... lrwxrwxrwx root root
snap 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
spool 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
tmp 4,096 File folder 17-02-2026 12:... drwxrwxrwt root root
www 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
.updated 208 UPDATED File 12-12-2025 03:... -rw-r--r-- root root


Remote files
Filter:
/var/www
Name | Size | Type | Date Modified | Permissions | Owner | Group
html 4,096 File folder 17-02-2026 12:... drwxrwxrwx root root

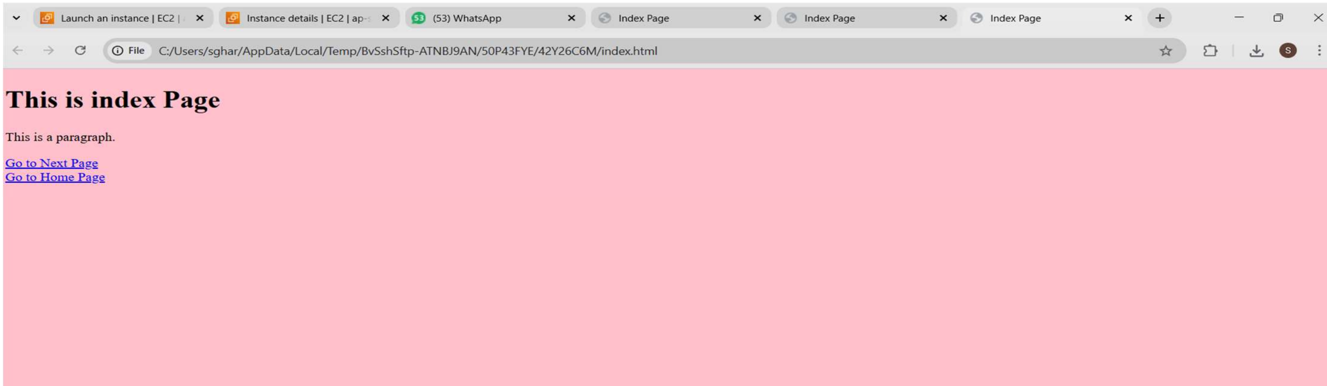

Local files
Filter:
C:\Users\sghar\OneDrive\Desktop\aws file
Name | Size | Type | Date Modified | Attributes
home.html 207 Chrome HT... 17-02-2026 12:... A
index.html 286 Chrome HT... 17-02-2026 12:... A
next.html 207 Chrome HT... 17-02-2026 12:... A
Remote files
Filter:
/var/www/html
Name | Size | Type | Date Modified | Permissions | Owner | Group
home.html 207 Chrome HT... 17-02-2026 12:... -rw-rw-r-- na na
index.html 286 Chrome HT... 17-02-2026 12:... -rw-rw-r-- na na
index.nginx-debian.html 615 Chrome HT... 17-02-2026 12:... -rw-r--r-- root root
next.html 207 Chrome HT... 17-02-2026 12:... -rw-rw-r-- na na

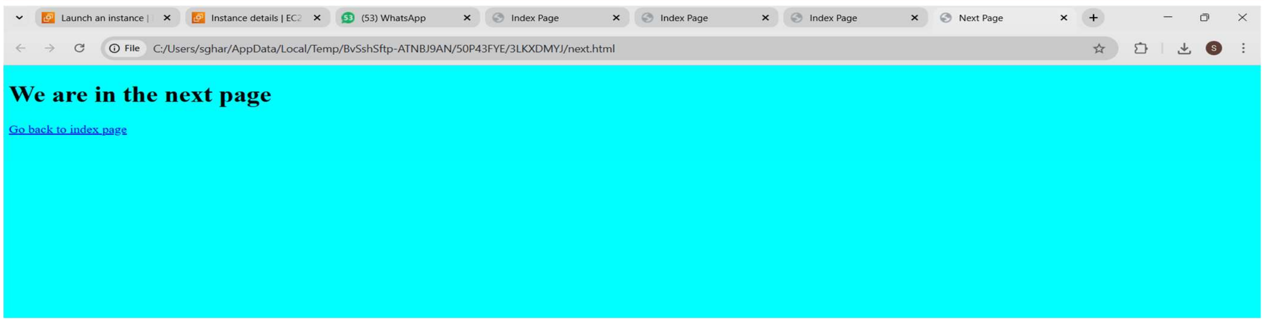
Step 11: Access the Website

Open a new browser tab and enter:

http://<Public-IPv4-address>

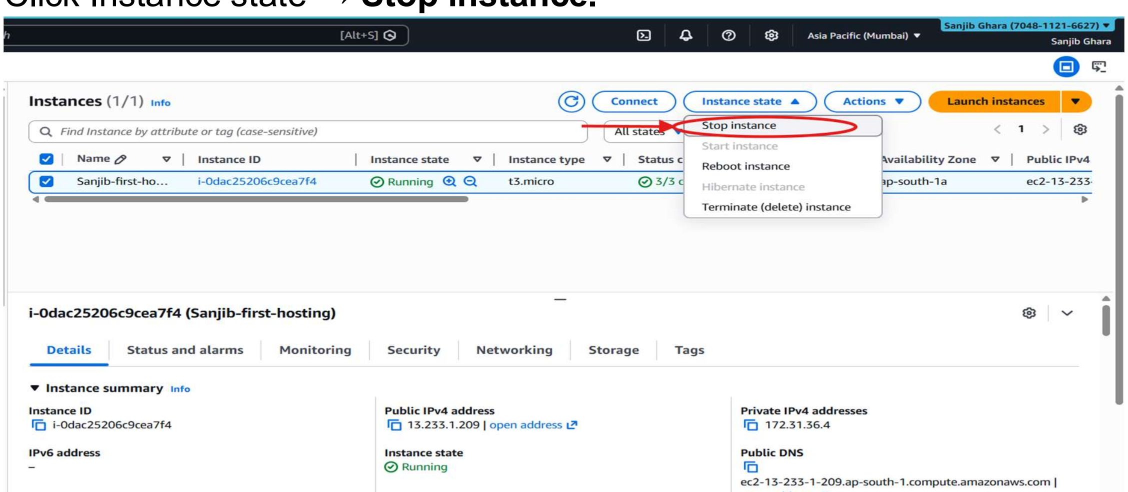
The static website will load successfully.



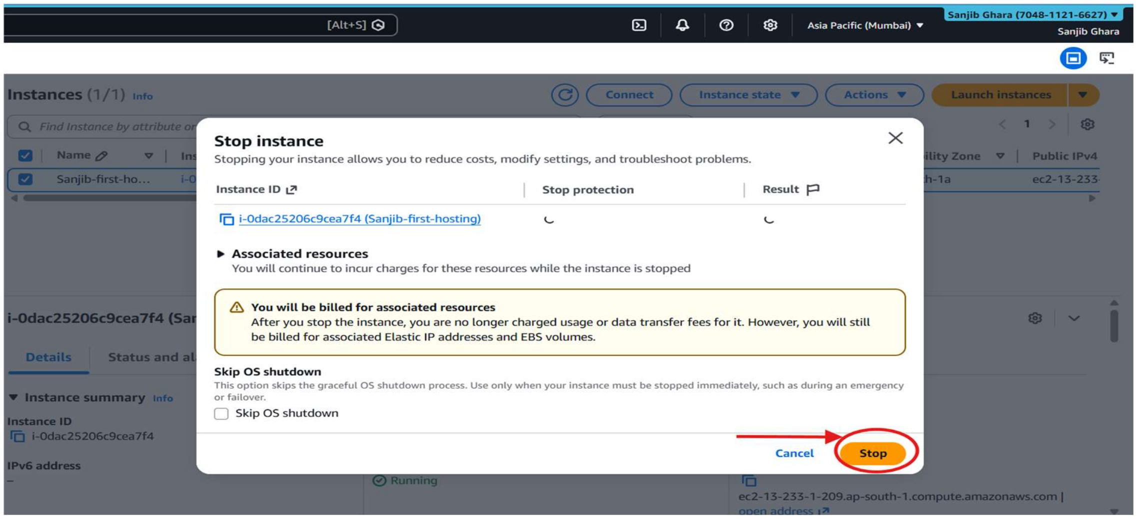


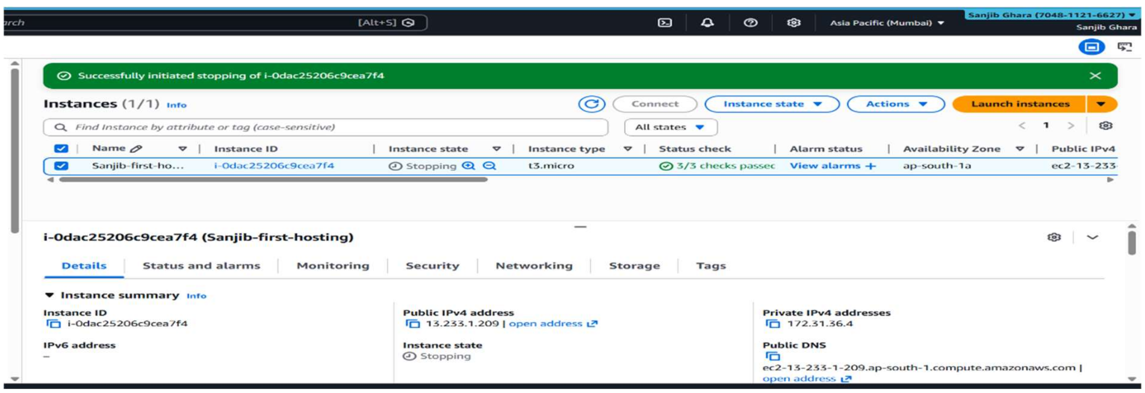
Step 12: Stop the EC2 Instance

1. Go back to the EC2 Instances page.
2. Select the instance.
3. Click Instance state → **Stop instance.**



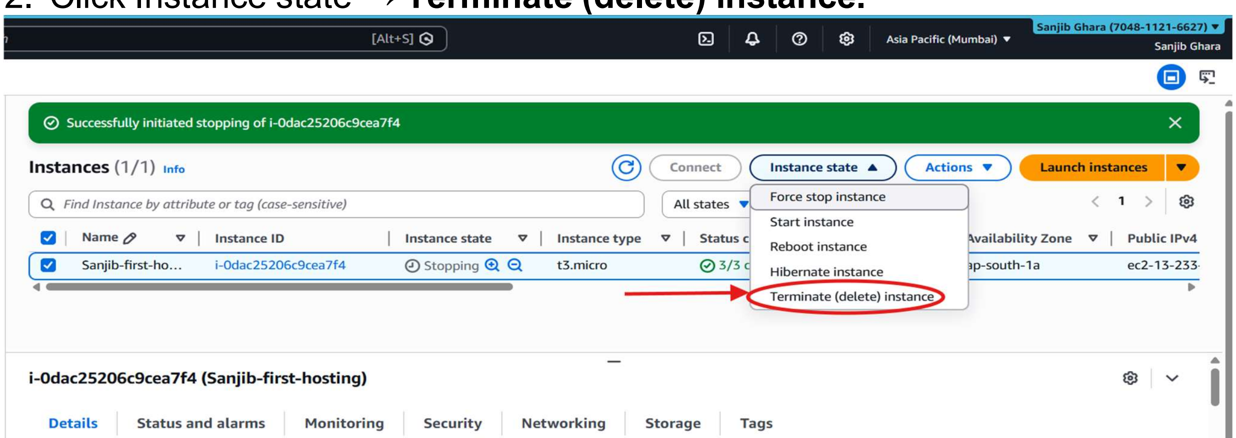
4. Confirm by clicking Stop.



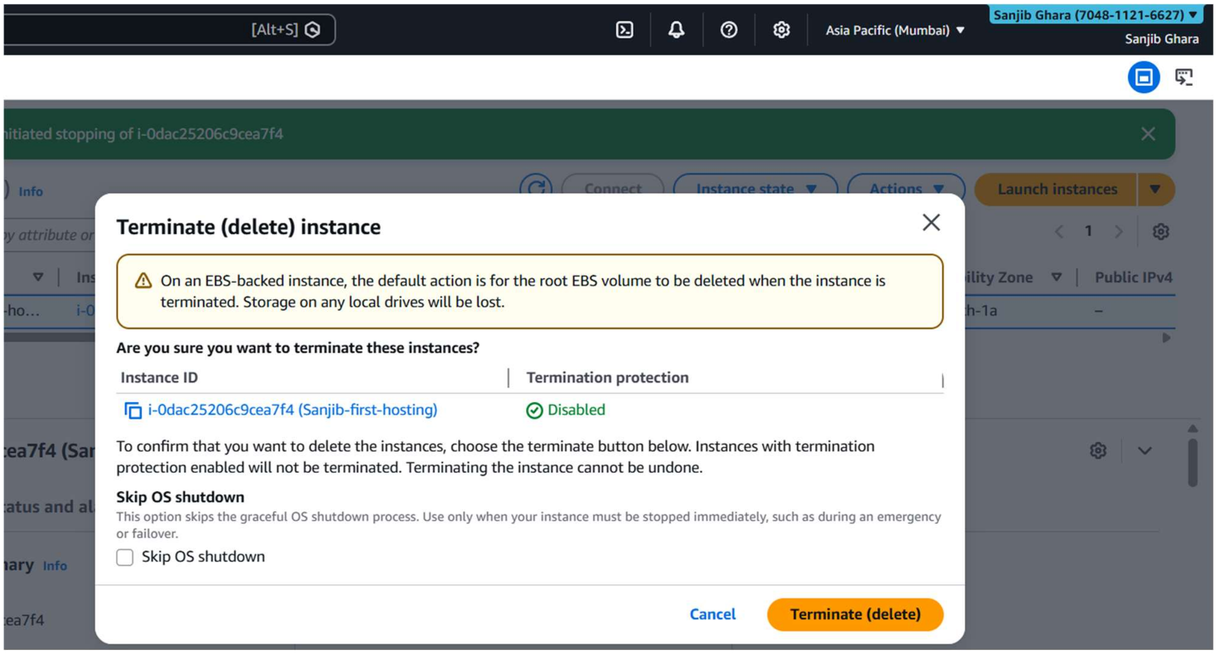


Step 13: Terminate the EC2 Instance

1. Select the same instance again.
2. Click Instance state → **Terminate (delete) instance.**

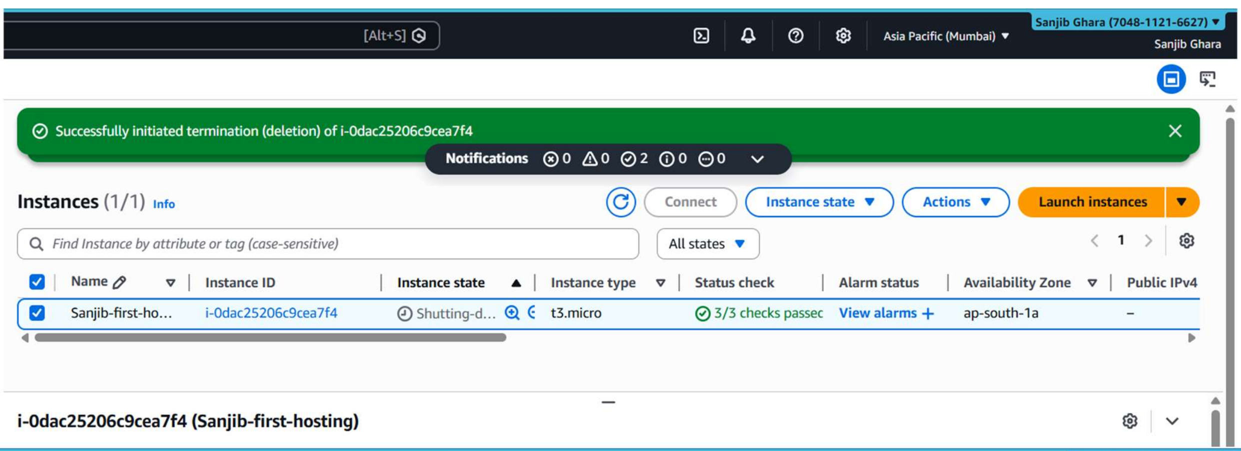


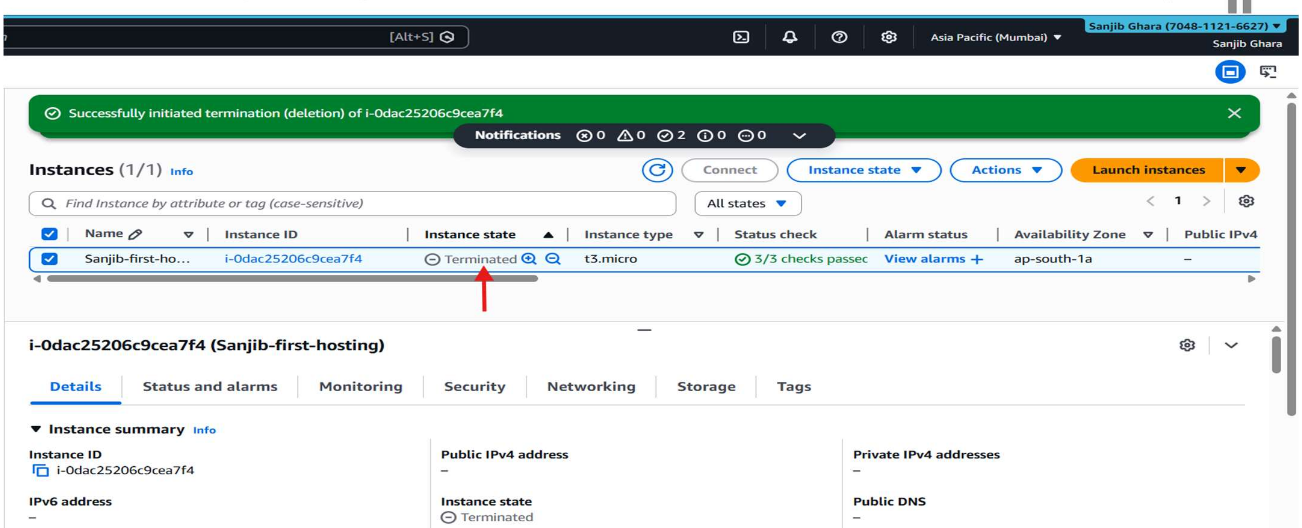
3. Confirm by clicking Terminate (delete).



Step 14: Instance Termination Confirmation

The EC2 instance is successfully terminated and all associated compute resources are released.





Conclusion

A static website consisting of multiple HTML files was successfully hosted on an Ubuntu-based EC2 instance using the Nginx web server and accessed via a public IPv4 address.